Diagonal Difference Task – Comparison between Recursive and Non-Recursive Approaches

Criteria	Non-Recursive Approach	Recursive Approach
Method	Uses iteration with loops	Uses recursion with function calls
Time Complexity	O(n)	O(n)
Space Complexity	O(1) constant space	O(n) due to call stack
Performance	Generally faster because it avoids recursive calls	Slightly slower due to recursive overhead
Code Structure	Simple and direct	More conceptual but harder to trace

Short Explanation:

The non-recursive solution calculates the sums of the two diagonals using a loop that iterates through the matrix once. This approach is memory efficient because it uses constant space. On the other hand, the recursive solution computes the diagonal sums by repeatedly calling a function for each matrix index. Although both approaches have the same time complexity O(n), the recursive approach consumes more memory due to the recursive call stack. Therefore, the non-recursive approach is generally more efficient in terms of memory usage and performance.